

PU5558-PU5567-computer-lab-report

Romina-Degan

Introduction

The following report is based on Provisional Patient Reported Outcome Measures (PROMS) specifically on the outcome of Hip and Knee Replacement procedures completed between April 2020 to March 2021, provided by NHS England. For this report the focus will be on the outcome measures of knee replacement surgery. Specific outcome measures pertaining to the Quality Of Life (QOL) to the patient have been recorded through a visual analogue scale designed to measure the self-reported status of the patients quality of life pre/post surgery. These values gathered from the survey will be used to inform the models.

Load necessary packages

```
library(tidyverse)
library(tidymodels)
library(corrplot)
library(psych)
library(ggplot2)
library(rmarkdown)
library(reshape2)
library(vip)
library(Hmisc)
library(VIM)
library(caret)
library(flextable)
library(gtsummary)
library(randomForest)
```

Load chosen dataset

```
kneeData <- read_csv("Knee_Replacement_CCG 2021.csv")
kneeData <- kneeData %>%
  select(-Procedure, -Year, -"Revision Flag", -"Provider Code") %>%
  drop_na() %>%
  unique()
```

```
glimpse(kneeData)
```

```
## Rows: 4,528
## Columns: 77
## $ `Age Band` <chr> "*", "*", "*", "*",~
## $ Gender <chr> "*", "*", "*", "*",~
## $ `Pre-Op Q Assisted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Assisted By` <dbl> 0, 0, 0, 0, 0, 0, 0~
## $ `Pre-Op Q Symptom Period` <dbl> 2, 4, 4, 2, 4, 2, 2~
## $ `Pre-Op Q Previous Surgery` <dbl> 2, 1, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Living Arrangements` <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Disability` <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Heart Disease` <dbl> 9, 9, 9, 9, 9, 1, 9~
## $ `High Bp` <dbl> 1, 9, 9, 1, 1, 1, 9~
## $ Stroke <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Circulation <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Lung Disease` <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ Diabetes <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ `Kidney Disease` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Nervous System` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Liver Disease` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Cancer <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Depression <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Arthritis <dbl> 9, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Mobility` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Self-Care` <dbl> 1, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Activity` <dbl> 3, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Discomfort` <dbl> 2, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Anxiety` <dbl> 1, 1, 2, 2, 1, 2, 1~
## $ `Pre-Op Q EQ5D Index Profile` <dbl> 21321, 21221, 21332~
## $ `Pre-Op Q EQ5D Index` <dbl> 0.364, 0.691, 0.030~
## $ `Post-Op Q Assisted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Assisted By` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Post-Op Q Living Arrangements` <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Post-Op Q Disability` <dbl> 1, 2, 2, 2, 2, 1, 2~
## $ `Post-Op Q Mobility` <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Self-Care` <dbl> 2, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Activity` <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Discomfort` <dbl> 1, 1, 1, 1, 1, 2, 2~
## $ `Post-Op Q Anxiety` <dbl> 2, 1, 1, 1, 1, 1, 1~
## $ `Post-Op Q Satisfaction` <dbl> 2, 3, 1, 2, 1, 4, 3~
## $ `Post-Op Q Success` <dbl> 1, 1, 1, 1, 1, 4, 1~
## $ `Post-Op Q Allergy` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Bleeding` <dbl> 2, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Wound` <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Urine` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Further Surgery` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Readmitted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q EQ5D Index Profile` <dbl> 12112, 11111, 11111~
## $ `Post-Op Q EQ5D Index` <dbl> 0.744, 1.000, 1.000~
## $ `Knee Replacement EQ 5D Index Post-Op Q Predicted` <dbl> 0.6621424, 0.740953~
## $ `Pre-Op Q EQ VAS` <dbl> 85, 50, 46, 60, 60,~
## $ `Post-Op Q EQ VAS` <dbl> 60, 100, 90, 95, 90~
## $ `Knee Replacement EQ VAS_Post-Op Q Predicted` <dbl> 75.63073, 66.38606,~
```

```
## $ `Knee Replacement Pre-Op Q Pain` <dbl> 1, 1, 0, 0, 1, 1, 1~
## $ `Knee Replacement Pre-Op Q Night Pain` <dbl> 1, 2, 2, 2, 2, 3, 0~
## $ `Knee Replacement Pre-Op Q Washing` <dbl> 3, 4, 3, 4, 3, 3, 3~
## $ `Knee Replacement Pre-Op Q Transport` <dbl> 1, 4, 2, 4, 2, 2, 2~
## $ `Knee Replacement Pre-Op Q Walking` <dbl> 3, 3, 2, 3, 3, 1, 2~
## $ `Knee Replacement Pre-Op Q Standing` <dbl> 3, 2, 2, 3, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Limping` <dbl> 0, 3, 0, 0, 3, 1, 1~
## $ `Knee Replacement Pre-Op Q Kneeling` <dbl> 0, 0, 0, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Work` <dbl> 2, 2, 1, 3, 2, 1, 2~
## $ `Knee Replacement Pre-Op Q Confidence` <dbl> 3, 3, 3, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Shopping` <dbl> 2, 4, 0, 2, 3, 0, 2~
## $ `Knee Replacement Pre-Op Q Stairs` <dbl> 1, 4, 2, 2, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Score` <dbl> 20, 32, 17, 27, 27, ~
## $ `Knee Replacement Post-Op Q Pain` <dbl> 4, 3, 3, 4, 3, 1, 2~
## $ `Knee Replacement Post-Op Q Night Pain` <dbl> 4, 4, 3, 4, 4, 3, 2~
## $ `Knee Replacement Post-Op Q Washing` <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Transport` <dbl> 2, 4, 4, 4, 4, 3, 4~
## $ `Knee Replacement Post-Op Q Walking` <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Standing` <dbl> 3, 4, 4, 4, 4, 1, 3~
## $ `Knee Replacement Post-Op Q Limping` <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Kneeling` <dbl> 0, 4, 0, 3, 2, 0, 2~
## $ `Knee Replacement Post-Op Q Work` <dbl> 3, 3, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Confidence` <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Shopping` <dbl> 4, 4, 4, 4, 4, 0, 4~
## $ `Knee Replacement Post-Op Q Stairs` <dbl> 3, 4, 4, 4, 3, 1, 4~
## $ `Knee Replacement Post-Op Q Score` <dbl> 37, 46, 42, 47, 44, ~
## $ `Knee Replacement OKS Post-Op Q Predicted` <dbl> 34.02619, 36.88715, ~
```

```
# glimpseData <- kneeData[, c("Age Band", "Gender", "Pre-Op Q Assisted", "Heart Disease", "Arthritis",
# "Post-Op Q Bleeding", "Pre-Op Q EQ5D Index Profile", "Post-Op Q EQ5D Index Profile",
# "Pre-Op Q EQ5D Index", "Post-Op Q EQ5D Index")]
# kable(glimpseData, format = "latex", booktabs = TRUE, longtable = TRUE,
# caption="Rate of Change within an Individual Health Board Against its Moving Average") %>%
# kable_styling(latex_options = c("striped", "hold_position")) %>%
# row_spec(0, bold = TRUE) %>%
# column_spec(1, "2cm") %>%
# row_spec(15, bold = TRUE) %>%
# kableExtra::landscape()

# set_flextable_defaults(
# font.family = "Arial", font.size = 10,
# border.color = "gray", big.mark = ""
# )

# ft <- flextable(head(kneeData), col_keys = c(
# "Age Band", "Gender", "Pre-Op Q Assisted", "Heart Disease", "Arthritis",
# "Post-Op Q Bleeding", "Pre-Op Q EQ5D Index Profile", "Post-Op Q EQ5D Index Profile",
# "Pre-Op Q EQ5D Index", "Post-Op Q EQ5D Index"
# )) |> bold(part = "header")
# set_table_properties(ft, width=.5, layout = "autofit")
```

Characteristic	N = 4,528 ¹
Age Band	
*	1,603 (35%)
50 to 59	164 (3.6%)
60 to 69	1,034 (23%)
70 to 79	1,511 (33%)
80 to 89	216 (4.8%)
Gender	
*	1,603 (35%)
1	1,326 (29%)
2	1,599 (35%)
Knee Replacement Post-Op Q Pain	
0	116 (2.6%)
1	773 (17%)
2	974 (22%)
3	1,529 (34%)
4	1,136 (25%)
Post-Op Q EQ5D Index	0.76 (0.69, 1.00)
Pre-Op Q Previous Surgery	
1	437 (9.7%)
2	4,060 (90%)
9	31 (0.7%)

¹n (%); Median (Q1, Q3)

Dataset description

This dataset uses data from NHS England regarding PROMS for knee replacement surgery. The way in which this data has been gathered is to collect self-reported symptoms/QOL metrics through two standard procedures. One survey is the EQ-5DTM Index designed to evaluate the QOL in terms of of doing daily tasks, such as (cite NHS PROMS): * Ability to pursue usual activities * Current experience of anxiety and/or depression, if any * Current experience of pain and discomfort if any * Mobility * Ability to wash and dress themselves. The closer the EQ5 value is to 1, the better the patients QOL.

While the EQ5 focuses on day to day tasks the EQ-VAS focuses on pre/post operative outcomes through a visual thermometer scale ranging from 0 to 100. This scale is designed to aid patients in describing how they feel post operation about the results and if their problems pre operation have been addressed with the surgery.

Visually inspecting the database we see that Age Band and Gender have many missing values. Alongside this, there are many missing values for co-morbidities. As instructed all missing values for co-morbidities denotes as 9 will be taken as a no value. However, since Age Band and Gender typically are important factors (especially as confounders), these values will be imputed. Since there are a possible 77 variables to provide summary statistics on, only specific columns relevant to the following questions below will be considered.

```
kneeData |> tbl_summary(include=c("Age Band", "Gender", "Knee Replacement Post-Op Q Pain", "Post-Op Q EQ5D
```

Suitable machine learning algorithm for three questions:

1. Before the operation, can we estimate the post-operative EQ5D index for a patient? The EQ5 index is provided as a float value between 0 and 1. Due to its continuous nature a suitable supervised model to answer this question would be a linear regression model. Since we wish to predict a continuous variable against possible predictor variables pre/post surgery, we would have to find correlations between an increasing EQ5 index value and variables in the dataset. There are 2 possible approaches to answering this question: a simple linear regression model and a random forest for regression. Both models won't be made, however assumptions on the dataset will guide the choice of model; if the dataset is found to have a non-linear relationship or standard test to check for normality does not hold true, a random forest will be used. A linear regression model is simpler to run and provides better interpretability of the model, however if assumptions do not hold, a random forest for regression must be used. This question will be used to complete the report.

2. Before the operation, can we predict how much pain a patient will have after the operation? Since pain is measured as ordinal categories, a model designed for classification would fit this specification. Referencing (mention table figure) we see that 25% of data points are missing values for the questions regarding pain. If we were to use an unsupervised model it may not capture the true relationship between predictors and the outcome variable due to the vast number of unlabelled data points for a significant predictor. To tackle this issue of missing data, a semi-supervised model such as a semi-supervised support vector machine can overcome the missing data barrier. The semi-supervised SVM would use a mixture of labelled and unlabelled data to create decision boundaries for the different pain categories, and predict the positioning of a data point to its relative class given the presence of certain predictor variables. The level of computation a semi-supervised SVM has may be detrimental to the model's overall performance, but coupled with the fact that there are many other missing values in other questions regarding the QOL status of the patient, imputation on specific questions would be difficult to complete without clinical knowledge to inform the possible current state of the patient. Therefore, the increased computational power for the model may be beneficial when trying to address the gaps in data.

3. Before the operation, can we calculate how many patients have had previous surgery?

Here we can use supervised/semi-supervised learning methods to calculate the numerical value of the patients that have had the previous surgery. The results are binary whether they have or have not had it.

Model building to answer chosen question

1. Data splitting

Before the data is split into relevant train/test splits. The issue of Age Band and Gender both missing 35% of their data needs to be handled. The following imputation is simple and doesn't use external questions or factors to inform decision making. Age Band is recoded into group numbers and the median of the group numbers is taken. Then the median value is then imputed to the missing data.

Imputation of the Gender column is slightly more complex, with the method being a k Nearest Neighbour (kNN) approach. This approach tries to maintain the randomness of the gender data points, (reference table) looking at the table there are 6% more males that get the surgery than females. Therefore using a simple 50-50 or a median split would lose the close barrier there is between the different genders. The following table summarises the current data points within the dataset

```
#Recode age bands into numeric groupings for an easier interpretation of the median
kneeData["Age Band"][kneeData["Age Band"] == "50 to 59"] <- "1"
kneeData["Age Band"][kneeData["Age Band"] == "60 to 69"] <- "2"
kneeData["Age Band"][kneeData["Age Band"] == "70 to 79"] <- "3"
kneeData["Age Band"][kneeData["Age Band"] == "80 to 89"] <- "4"
kneeData$"Age Band" <- as.numeric(as.character(kneeData$"Age Band"))

#Impute median values into the missing age data
kneeData["Age Band"][is.na(kneeData["Age Band"])] <- median(kneeData$"Age Band", na.rm = TRUE)
```

Characteristic	N = 4,528 ^I
Age Band	
1	164 (3.6%)
2	1,034 (23%)
3	3,114 (69%)
4	216 (4.8%)
Gender	
1	2,019 (45%)
2	2,509 (55%)

^I n (%)

```
kneeData[kneeData == "*"] <- NA

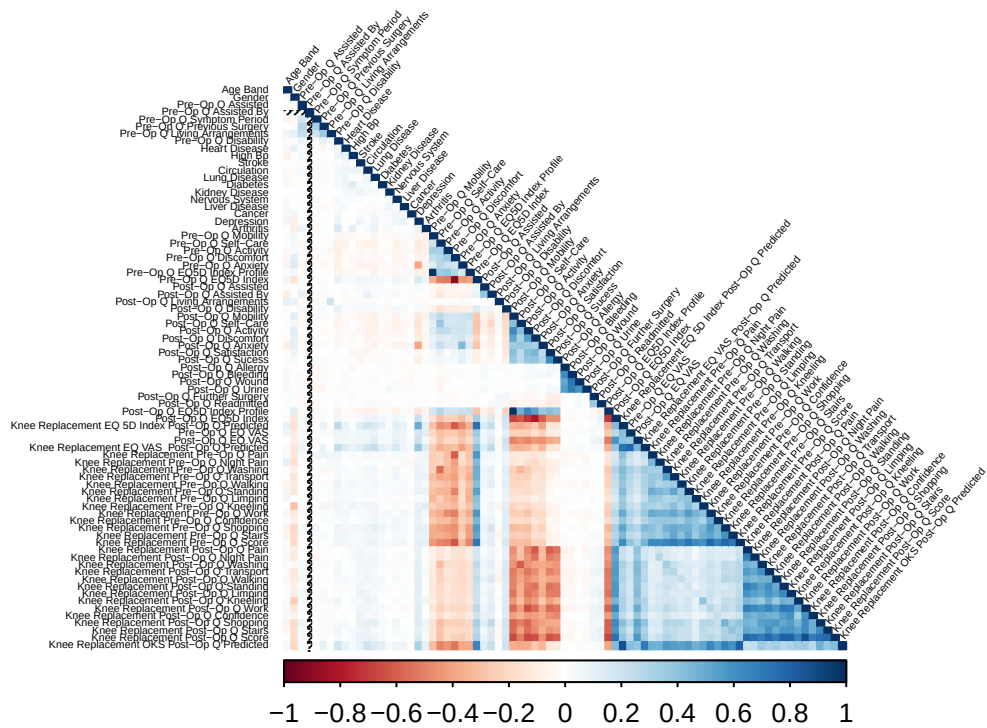
#Call a KNN function designed to split the
imputedGender <- kNN(kneeData, k = 1)
kneeData$Gender <- imputedGender$Gender
kneeData$Gender <- as.numeric(kneeData$Gender)

kneeData |> tbl_summary(include=c("Age Band","Gender" ))
```

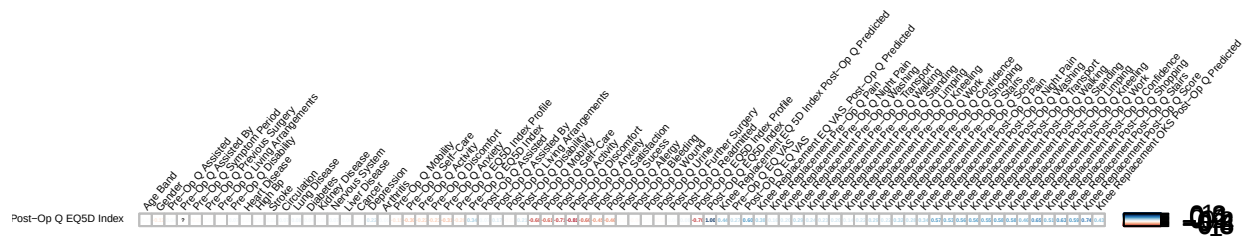
```
kneeSplit <- kneeData %>%
  initial_split(prop = 0.7, strata = "Post-Op Q EQ5D Index")
kneeTrain <- training(kneeSplit)
kneeTest <- testing(kneeSplit)
```

2. Selection and preprocessing of predictors

```
kneeCorr <- cor(kneeTrain)
corrplot(kneeCorr, method = "color", type = "lower", tl.col = "black", tl.srt = 50, number.cex = 0.5, t
```



```
kneeCorrPostOp <- t(kneeCorr[, c(46), drop = FALSE])
corrplot(kneeCorrPostOp, method = "number", tl.col = "black", tl.srt = 50, tl.cex = 0.3, cex.axis=0.3, num
```



```
for (i in seq_along(kneeTrain)) {
  cat("Index:", i, "- Name:", colnames(kneeTrain)[i], "\n")
}
```

```
## Index: 1 - Name: Age Band
## Index: 2 - Name: Gender
## Index: 3 - Name: Pre-Op Q Assisted
## Index: 4 - Name: Pre-Op Q Assisted By
## Index: 5 - Name: Pre-Op Q Symptom Period
## Index: 6 - Name: Pre-Op Q Previous Surgery
## Index: 7 - Name: Pre-Op Q Living Arrangements
## Index: 8 - Name: Pre-Op Q Disability
## Index: 9 - Name: Heart Disease
## Index: 10 - Name: High Bp
## Index: 11 - Name: Stroke
## Index: 12 - Name: Circulation
## Index: 13 - Name: Lung Disease
## Index: 14 - Name: Diabetes
## Index: 15 - Name: Kidney Disease
## Index: 16 - Name: Nervous System
## Index: 17 - Name: Liver Disease
## Index: 18 - Name: Cancer
## Index: 19 - Name: Depression
## Index: 20 - Name: Arthritis
## Index: 21 - Name: Pre-Op Q Mobility
## Index: 22 - Name: Pre-Op Q Self-Care
```

Index: 23 - Name: Pre-Op Q Activity
 ## Index: 24 - Name: Pre-Op Q Discomfort
 ## Index: 25 - Name: Pre-Op Q Anxiety
 ## Index: 26 - Name: Pre-Op Q EQ5D Index Profile
 ## Index: 27 - Name: Pre-Op Q EQ5D Index
 ## Index: 28 - Name: Post-Op Q Assisted
 ## Index: 29 - Name: Post-Op Q Assisted By
 ## Index: 30 - Name: Post-Op Q Living Arrangements
 ## Index: 31 - Name: Post-Op Q Disability
 ## Index: 32 - Name: Post-Op Q Mobility
 ## Index: 33 - Name: Post-Op Q Self-Care
 ## Index: 34 - Name: Post-Op Q Activity
 ## Index: 35 - Name: Post-Op Q Discomfort
 ## Index: 36 - Name: Post-Op Q Anxiety
 ## Index: 37 - Name: Post-Op Q Satisfaction
 ## Index: 38 - Name: Post-Op Q Success
 ## Index: 39 - Name: Post-Op Q Allergy
 ## Index: 40 - Name: Post-Op Q Bleeding
 ## Index: 41 - Name: Post-Op Q Wound
 ## Index: 42 - Name: Post-Op Q Urine
 ## Index: 43 - Name: Post-Op Q Further Surgery
 ## Index: 44 - Name: Post-Op Q Readmitted
 ## Index: 45 - Name: Post-Op Q EQ5D Index Profile
 ## Index: 46 - Name: Post-Op Q EQ5D Index
 ## Index: 47 - Name: Knee Replacement EQ 5D Index Post-Op Q Predicted
 ## Index: 48 - Name: Pre-Op Q EQ VAS
 ## Index: 49 - Name: Post-Op Q EQ VAS
 ## Index: 50 - Name: Knee Replacement EQ VAS_Post-Op Q Predicted
 ## Index: 51 - Name: Knee Replacement Pre-Op Q Pain
 ## Index: 52 - Name: Knee Replacement Pre-Op Q Night Pain
 ## Index: 53 - Name: Knee Replacement Pre-Op Q Washing
 ## Index: 54 - Name: Knee Replacement Pre-Op Q Transport
 ## Index: 55 - Name: Knee Replacement Pre-Op Q Walking
 ## Index: 56 - Name: Knee Replacement Pre-Op Q Standing
 ## Index: 57 - Name: Knee Replacement Pre-Op Q Limping
 ## Index: 58 - Name: Knee Replacement Pre-Op Q Kneeling
 ## Index: 59 - Name: Knee Replacement Pre-Op Q Work
 ## Index: 60 - Name: Knee Replacement Pre-Op Q Confidence
 ## Index: 61 - Name: Knee Replacement Pre-Op Q Shopping
 ## Index: 62 - Name: Knee Replacement Pre-Op Q Stairs
 ## Index: 63 - Name: Knee Replacement Pre-Op Q Score
 ## Index: 64 - Name: Knee Replacement Post-Op Q Pain
 ## Index: 65 - Name: Knee Replacement Post-Op Q Night Pain
 ## Index: 66 - Name: Knee Replacement Post-Op Q Washing
 ## Index: 67 - Name: Knee Replacement Post-Op Q Transport
 ## Index: 68 - Name: Knee Replacement Post-Op Q Walking
 ## Index: 69 - Name: Knee Replacement Post-Op Q Standing
 ## Index: 70 - Name: Knee Replacement Post-Op Q Limping
 ## Index: 71 - Name: Knee Replacement Post-Op Q Kneeling
 ## Index: 72 - Name: Knee Replacement Post-Op Q Work
 ## Index: 73 - Name: Knee Replacement Post-Op Q Confidence
 ## Index: 74 - Name: Knee Replacement Post-Op Q Shopping
 ## Index: 75 - Name: Knee Replacement Post-Op Q Stairs
 ## Index: 76 - Name: Knee Replacement Post-Op Q Score

```
## Index: 77 - Name: Knee Replacement OKS Post-Op Q Predicted
```

The full correlation plot is displayed above but the values and their labels are difficult to read with the 77 variables that we have to consider. Extrcation of significant correlations in the main plot will be used to hilight the variables that will be used as predictors for the model. Correlation values were observed through the plots and all predictor values that were above 0.5 or below -0.5 were considered as significant predictors. This is puerly numerically based, but given clinical expertise there may be some predictors that may want to be included into the final model.

```
colnames(kneeTrain) <- make.names(colnames(kneeTrain))
colnames(kneeTest) <- make.names(colnames(kneeTest))

kneeModelRec <- kneeTrain %>%
  recipe(Post.Op.Q.EQ5D.Index ~
    Post.Op.Q.Mobility +
    Post.Op.Q.Self.Care +
    Post.Op.Q.Discomfort +
    Post.Op.Q.EQ5D.Index.Profile +
    Post.Op.Q.EQ.VAS +
    Knee.Replacement.Post.Op.Q.Pain +
    Knee.Replacement.Post.Op.Q.Night.Pain +
    Knee.Replacement.Post.Op.Q.Washing +
    Knee.Replacement.Post.Op.Q.Transport +
    Knee.Replacement.Post.Op.Q.Walking +
    Knee.Replacement.Post.Op.Q.Standing +
    Knee.Replacement.Post.Op.Q.Limping +
    Knee.Replacement.Post.Op.Q.Work +
    Knee.Replacement.Post.Op.Q.Confidence +
    Knee.Replacement.Post.Op.Q.Shopping +
    Knee.Replacement.Post.Op.Q.Stairs +
    Knee.Replacement.Post.Op.Q.Score) %>%
  step_zv(all_predictors()) %>%
  step_corr(all_predictors())
```

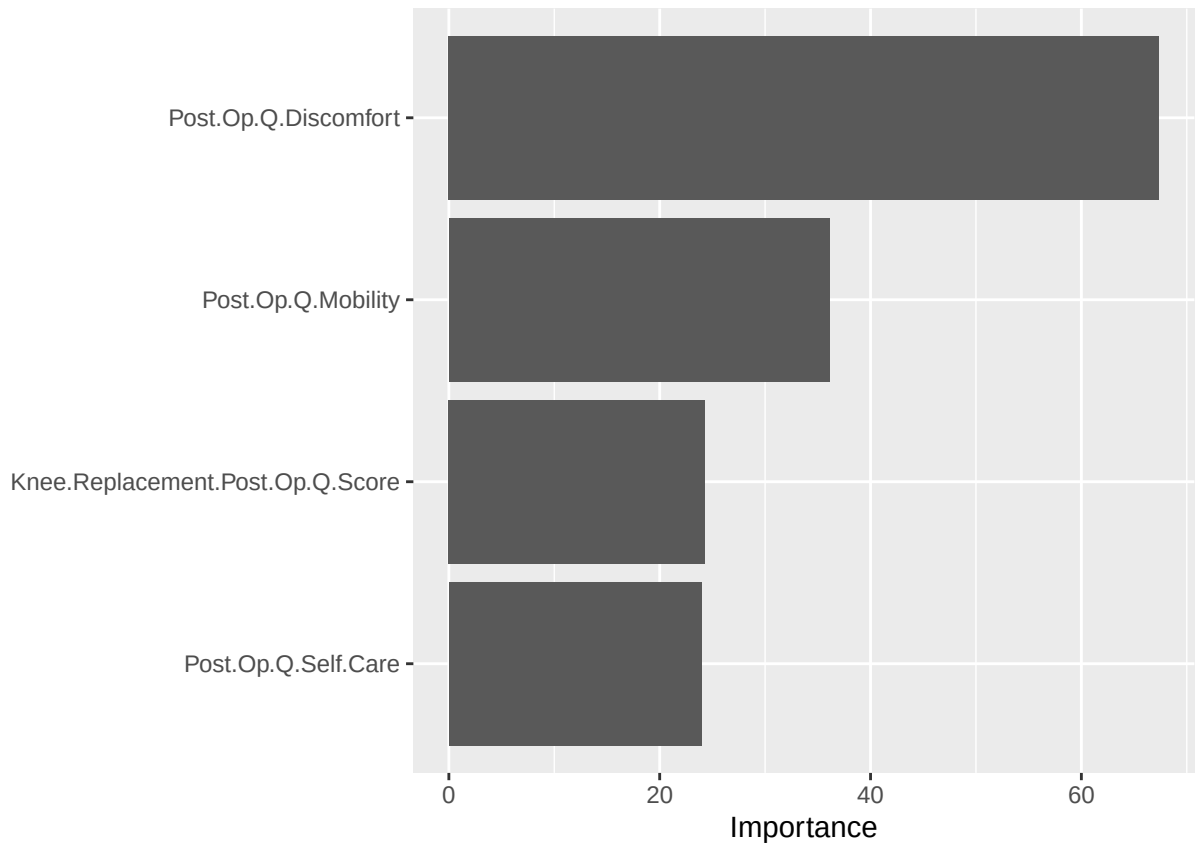
3. Model specification and training

```
kneeRandForest <- rand_forest() %>%
  set_mode("regression") %>%
  set_engine("randomForest")

kneeWorkflow <- workflow() %>%
  add_recipe(kneeModelRec) %>%
  add_model(kneeRandForest)

kneeFitModel <- kneeWorkflow %>%
  fit(data = kneeTrain)

kneeFitModel %>%
  extract_fit_parsnip() %>%
  vip(num_features=4)
```



4. Model evaluation Code has

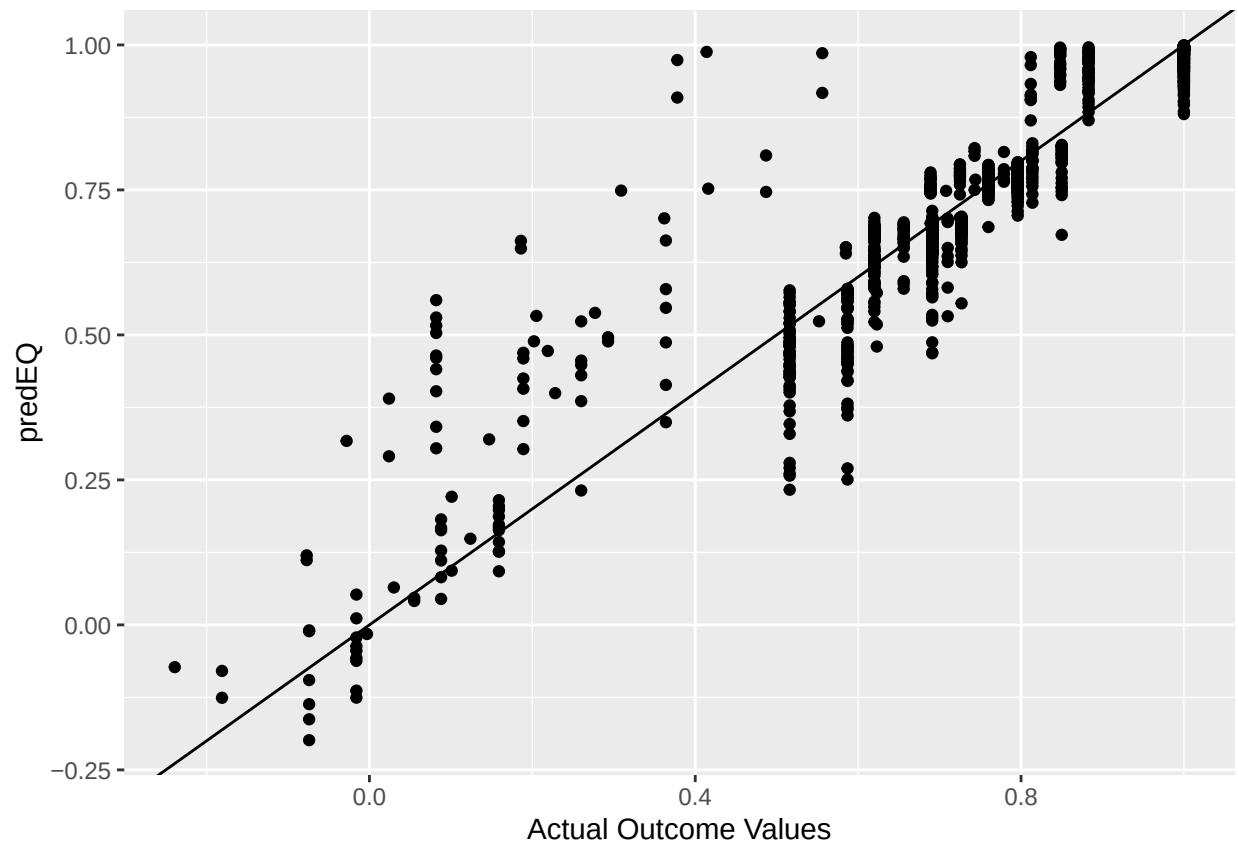
```
predEQVal <- kneeFitModel %>%
  predict(new_data = kneeTest)

kneeResult <- kneeTest %>%
  bind_cols(predEQVal) %>%
  rename(predEQ = .pred)

metrics(kneeResult, truth = Post.Op.Q.EQ5D.Index, estimate = predEQ)

## # A tibble: 3 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>      <dbl>
## 1 rmse    standard      0.0823
## 2 rsq     standard      0.869
## 3 mae     standard      0.0468

kneeResult %>%
  ggplot(aes(x = Post.Op.Q.EQ5D.Index, y = predEQ)) +
  geom_abline(intercept = 0, slope = 1) +
  geom_point() +
  xlab("Actual Outcome Values")
```



```
ylab("Predicted Outcome Values")
```

```
## $y  
## [1] "Predicted Outcome Values"  
##  
## attr(,"class")  
## [1] "labels"
```

Limitations of machine learning model